NEEDLE: A Live Benchmark of Web Search APIs for AI Agents

Anonymous ACL submission

Abstract

NEEDLE measures web search APIs on five tasks: news, finance, papers, law and rare-word queries. It draws queries and reference answers from public sources on each run, except for a fixed rare-query stream, and scores the returned titles, URLs and snippets. We study 26 API configurations. Across 12 days with all 21 main engines present, overall shares of a pooled ceiling span 15–73%; paired tests separate 178 of 210 engine pairs after Bonferroni adjustment. Title queries hide large differences: 23 configurations achieve scholar recall@10 of .71–.98 from titles, but .08–.72 from body keywords. Keenable leads all eight weekly comparisons, while Perplexity leads on a later day's runs. Blinded LLM raters from three vendors agree with the relevance judge at κ .77–.79; no human audited the scorers. The results support choosing an API by task and paired comparison, not by rank alone. Code, data and dashboard are public; links are withheld for review.

1 Introduction

An agent can read a paper only if its search API finds it. Over 30 days, Keenable found papers from body-keyword queries such as "gap rent regularity" with recall@10 .72; ChatGPT search reached .08. Both exceeded .80 on title queries (Figure 1a). The choice of query can therefore hide or expose a large retrieval gap.

NEEDLE tests the search API alone: one query in, up to k results out, scored on the title, URL and snippet an agent receives. It does not score the agent's final answer, as BrowseComp does (Wei et al., 2025), or hold the corpus fixed, as FreshStack does (Thakur et al., 2025). Fresh queries and reference answers limit dependence on a fixed test set, which can leak into training data (Sainz et al., 2023; Balloccu et al., 2024), though registry facts and rare queries can repeat.

The benchmark combines five tasks, automatic scoring and a pooled ceiling: the best results found by any tested configuration. We report:

- Results for 26 configurations, including paired tests on 12.4k query-runs shared by the 21 main engines.
- Paired scholar query types that expose differences between title and full-text retrieval.
- Checks of the judge, matchers, ceiling and failure rules, with samples and labels for readers to inspect.

2 Related Work

Live benchmarks. LiveBench, LiveCodeBench, RealTime QA, FreshQA, LiveSearchBench and LiveNewsBench draw new items from dated sources (White et al., 2025; Jain et al., 2025; Kasai et al., 2023; Vu et al., 2024; Zhou et al., 2026a; Zhang et al., 2026). They score a model's answer; FreshStack scores retrievers on a fixed corpus (Thakur et al., 2025).

Agents and search APIs. BrowseComp, GAIA and AssistantBench score agents end to end (Wei et al., 2025; Mialon et al., 2024; Yoran et al., 2024); BrowseComp-Plus fixes the corpus to isolate the retriever (Chen et al., 2026). Recent studies and leaderboards swap the search API under one agent on static questions (Selvam and Ghosh, 2026; Artificial Analysis, 2026; Openbenchmarks, 2026). Other studies describe how agents search (Ning et al., 2026; Pezzuti et al., 2026). Web engine audits compare engines on fixed query sets (Lewandowski, 2015; Bevendorff et al., 2024; Kirsten et al., 2026).

Judges and known items. LLM assessors can reproduce system rankings (Thomas et al., 2024; Upadhyay et al., 2025; Arabzadeh and Clarke, 2025). Simulated known-item and tip-of-tongue queries can rank systems (Azzopardi et al., 2007; He et al., 2025), and full text helps retrieval beyond

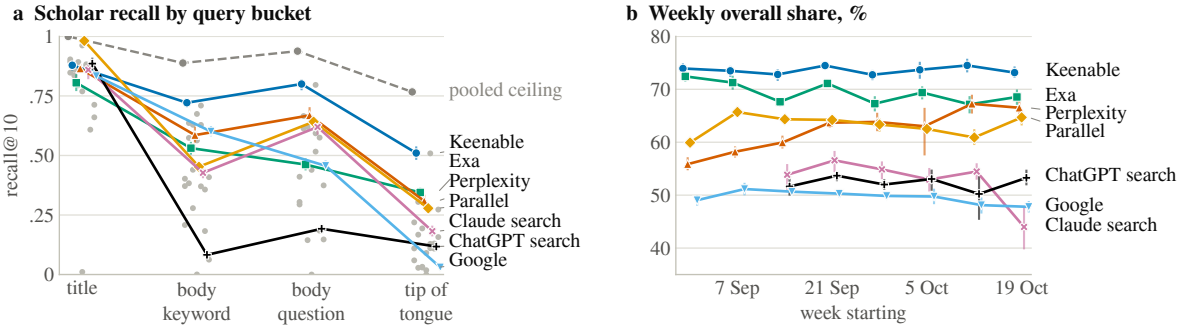


Figure 1: (a) Scholar recall@10 by query type: 31 runs, 29 September–28 October 2026; 2,600–3,005 query-runs per line and type. Lines show the seven engines in (b), grey dots the other 19 configurations, and dashes the pooled ceiling. (b) Overall share by ISO week, 31 August–25 October 2026. ChatGPT and Claude search start on 14 September; the pool grows from 18 to 25 configurations. Points are query-weighted means; bars are 95% run-level bootstrap intervals. Scores follow the dashboard rules.

abstracts (Lin, 2009). Domain benchmarks cover literature, caselaw and finance search (Ajith et al., 2024; Hou et al., 2025; Hu et al., 2026). NEEDLE combines fresh queries, isolated API scoring and these scorers.

3 The Benchmark

A *configuration* is an API at one setting. We test 26 configurations from 20 vendors, including five mode variants, and call each an *engine*. A *query-run* is one query issued in one run. Appendix A gives the workflow, request settings and run counts.

3.1 Queries

News. Each hour an LLM writes short event queries from the newest items in 124 feeds and Google Trends RSS for 52 US geographies. A run samples 20 queries across topics and rejects evergreen items. **Finance.** Queries ask for company facts from Wikidata (Vrandečić and Krötzsch, 2014) and GLEIF, quarterly 10-Q figures from SEC XBRL, or filings by description. **Scholar.** Each recent arXiv or Europe PMC (Europe PMC Consortium, 2015) paper yields four paired known-item queries (Azzopardi et al., 2007): its normalized *title*; *body keywords* for a detail absent from title and abstract; a *body question* about that detail; and a *tip-of-tongue* description (Arguello et al., 2021) without names or exact values. All four must pass the filters. **Legal.** Recent opinions from 14 federal courts on CourtListener yield caption queries. eCFR sections yield queries based on a verbatim two-to-eight-word span, quoted in one of four variants. Filters reject citation leaks.[1] **Rare.** We sample a fixed public stream of agent queries (Ning et al., 2026; Li et al., 2026; Zhou et al., 2026b), keeping English ones (fastText language ID; Joulin et al., 2017, 2016) that contain a word split into at least five BERT WordPieces (Wu et al., 2016; Devlin et al., 2019). Appendix B gives filters and examples.

At least half of legal queries and finance money and filing queries carry quotes, `site:`, `after:` or `before:`. The client maps operators to native filters where available (Appendix F).

3.2 Scores

News and rare. GPT-6 rates each distinct result from 0 to 4 using a shared query profile, with checks for relevance, coverage, freshness and literal operator compliance, as LLM relevance assessors do (Thomas et al., 2024; Upadhyay et al., 2025). The scale adapts the Needs Met levels of Google (2025). Ratings 0–4 map to gains $\{0, 0, .117, .667, 1\}$; this design choice puts high weight on strong results (Kanoulas and Aslam, 2009). Snippets have a 2,000-character cap. Repeated URLs score zero except on `site:` queries.

For each query, the *ceiling* keeps the best rating per normalized URL across all configurations, variants included, and sorts these ratings, as in pooled test collections (Spärck Jones and van Rijsbergen, 1975; Zobel, 1998). An engine's nDCG@5 (Järvelin and Kekäläinen, 2002) uses this list as its ideal. We drop a query-run from all engines if any judgment fails or no result reaches rating 2.

[1] Sources: `trends.google.com`, `api.gleif.org`, `data.sec.gov`, `courtlistener.com`, `ecfr.gov`.

Appendix B.3 gives the formula and an example.

Scholar, legal and finance. Recall@k counts queries with a matching result in the top k. Scholar matches arXiv ids, DOIs and PMIDs; legal matches citations, dockets and identifiers in URLs and snippets. Caselaw recall measures case identity, not necessarily the opinion page. Finance checks whether the snippets contain the answer, with a 0.1% money band and a 15% headcount band. It reads table figures and parenthesized losses. An LLM may upgrade a non-money miss, but cannot revoke a hit; a failed backstop judgment excludes an engine's query only if no result matches. The ceiling is a hit by any engine. Appendix E checks matcher error.

3.3 Overall share and failures

For bench metric M_b, an engine's overall share is

$$S(e) = \frac{1}{5}\sum_{b=1}^{5}\frac{M_b(e)}{M_b(\text{ceiling})}.$$

The benches have equal weight. The ceiling is the best the tested engines found, not the best the web contains.

Table 1 uses dashboard scores from the 276 runs in which all 21 main engines took part. On finance, scholar and legal, account-side failures (our quota, rate limit, login or credit) are excluded; other failures score zero. On news and rare, every failure in this window scores zero. These exclusions remove up to 28% of an engine's query-runs. Intervals use run-level bootstraps (Efron and Tibshirani, 1993) and measure sampling error, not scorer error. Appendices A and C specify the rules and test alternatives.

4 Results

Most engine pairs differ, but rank alone is not enough. Overall shares span 14.7% (Ceramic) to 73.2% (Keenable). Run-resampled paired tests separate 188 of 210 pairs at 95%, 178 after Bonferroni adjustment (Carterette, 2012), and 164 under both day-level resampling and matched query support. Only Keenable and Ceramic differ from every other engine after adjustment. Scholar supplies 3.7 points of Keenable's 4.7-point lead over Exa; without it the lead is 1.2 points. Appendix C reports the uncertainty and weighting checks.

Body queries expose gaps that title queries hide. Over 31 scholar runs, 23 of 26 configurations reach recall@10 .71–.98 from titles. The same 23 span .08–.72 on body keywords (Figure 1a; Table 8). Keenable leads at .72, followed by its realtime mode (.71) and Kagi (.67).

Weekly ranks are stable, not fixed. Adjacent weekly rankings agree at Kendall's τ_b (Kendall, 1945) .81–.94, and Keenable leads all eight weeks (Figure 1b). Lower ranks shift: Perplexity passes Parallel, and Claude search falls about 13 points, about half due to our credit lapse. A later day's runs put Perplexity ahead of Keenable, 72.4 to 71.3. That change rests on one scholar and one legal run, within each bench's run-to-run range (Appendix C).

LLM raters broadly agree with the judge. Blinded raters from three vendors grade 1,200 results with quadratic-weighted κ (Cohen, 1968) .90–.94 with one another and .77–.79 with the judge. On 100 complete query pools, two raters each rank engines at Spearman .94 with the judge, but neither separates Keenable from Exa. These are checks of model agreement, not human validation. Removing an engine from the ceiling changes its overall share by at most 0.8 points. Appendices D and C give the checks.

A site restriction can reduce recall. In a paired test, adding `site:courtlistener.com` to the same 80 case captions lowers mean recall@5 by .36, from .85; 15 of 18 engines lose beyond their 95% intervals. Site restrictions help SEC filings (+.35) and CFR sections (+.25). An earlier test with paired queries found that most operators had no significant effect on three web engines (Eastman and Jansen, 2003). The test does not separate index coverage, ranking and filter behavior (Appendix F.2).

Limitations

Conflict of interest. The authors' organization builds an evaluated engine. The code, prompts, clients and per-query outputs are public so readers can audit the claims. All engines receive the same queries, snippet cap, request pacing and judge; each keeps one setting throughout the main window. We cannot tell whether a vendor changed its index for the benchmark or for other reasons.

Evidence and judge. Scores measure the API response, not the fetched page. Short snippets can hide relevant content (Turpin et al., 2009; Kazai

Engine	Overall share, %		Scholar	News	Finance	Rare	Legal	Failed, %	Latency
	mean	95% CI	R@10	nDCG@5	R@5	nDCG@5	R@5	requests	p_{50}, ms
ceiling (pooled)	*100*		*.901*	*1*	*.993*	*1*	*.971*		
Keenable	73.2	72.3–74.1	**.731**	.609	.903	**.508**	**.797**	0.0	135
Exa	68.5	67.4–69.4	.563	.599	**.922**	.476	.771	0.3	2,132
Perplexity	66.9	65.3–68.5	.578	.640	.895	.468	.674	0.0	433
Parallel	64.9	63.6–66.1	.587	.535	.907	.387	.734	0.8	2,660
Brave (LLM ctx)	64.6	63.4–65.6	.533	**.642**	.858	.454	.658	0.0	1,112
Linkup	57.6	56.1–58.9	.470	.567	.797	.400	.570	0.0	1,212
Brave	57.3	56.2–58.4	.488	.557	.769	.354	.621	0.0	487
Context	57.1	55.4–58.6	.598	.469	.782	.339	.576	0.0	866
DeepSeek	56.9	55.8–57.9	.271	.635	.818	.387	.676	0.0	551
You.com	55.8	54.7–56.9	.471	.553	.747	.343	.602	0.0	486
ChatGPT search†§	51.5	49.4–53.1	.300	.340	.809	.366	.697	11.1	43,831
Kagi§	51.3	49.9–52.6	.560	.301	.728	.275	.617	7.6	1,275
Firecrawl	50.7	49.2–52.0	.488	.286	.820	.265	.598	0.0	688
Octen	50.2	48.8–51.7	.307	.511	.730	.322	.586	0.0	197
Google (Serper)	47.7	46.8–48.6	.475	.262	.738	.255	.582	0.0	1,785
Claude search†§	45.1	42.7–47.6	.498	.276	.685	.195	.525	35.9	33,906
TinyFish§	44.5	43.0–45.9	.429	.256	.730	.220	.521	29.0	1,904
Bing (SearchAPI)	44.1	41.1–46.8	.416	.288	.675	.173	.586	0.4	2,008
Jina	43.8	42.4–45.1	.492	.210	.726	.163	.523	4.9	17,272
Tavily	42.6	41.5–43.7	.303	.303	.828	.284	.361	0.1	1,999
Ceramic	14.7	13.9–15.7	.001	.040	.441	.109	.137	0.1	99

Table 1: Results on 276 shared runs, 17–28 October 2026 (12.4k query-runs, 21 engines); the ceiling also includes variants. Bench cells are absolute metrics, best in bold; italics mark the ceiling. Failed counts all request failures. Latency is the per-run median on news. †Server-side search tool; latency includes answer generation. §Account-side exclusions leave rank uncertain (Appendix C).

et al., 2022), and identifier matching can miss a target without its id. On 400 sampled results, page text changes 25% of grades. News and rare depend on one judge; all audit raters are LLMs, and none provides human validation, which LLM labels cannot replace (Faggioli et al., 2023; Clarke and Dietz, 2025). Shared rules can cause shared errors. Query-term stuffing can fool judges (Alaofi et al., 2024); vendor and model biases can affect scores (Balog et al., 2025; Soboroff, 2025). The judge shares a vendor with ChatGPT search, and a second judge shares one with Claude search; LLM evaluators can favor their own outputs (Panickssery et al., 2024).

Matcher error. Finance table figures can match the wrong period when snippets omit column headers. A fresh audit confirms only 70% of table-reading hits; most others are unresolved, not contradicted. The pooled error estimates cannot show each engine's error rate. Caselaw recall credits pages about the case, not just the opinion. Appendices D and E bound these effects.

Coverage and weights. Queries are English and US-centric. Equal bench weights do not estimate agent traffic, and the lead depends strongly on scholar. Default API tiers need not be each vendor's best. ChatGPT and Claude search return sources through answer-generating tools, so their quality scores are comparable but their latency and cost are not. The ceiling depends on the engine pool (Buckley et al., 2007); compare shares within a common window.

Failures and timing. Excluding account-side failures assumes those requests would score like the rest. Without that assumption, several middle ranks remain uncertain. News and rare also count our credit lapses as zeros in the main window. Engines process queries at different speeds, so slow engines see news later. A five-minute dispatch filter leaves the news order unchanged. Vendor redirect links can break URL matching in later runs.

Repetition and reproducibility. Registry facts and the fixed rare stream can repeat. Public sources and queries permit tuning or special handling of benchmark traffic. We archive queries, reference answers, results and judgments, but not API publication dates or each run's code revision. Stored ratings reproduce the tables; re-judging cannot exactly reproduce the inputs. Appendix A lists the release terms and reconstruction steps.

References

Anirudh Ajith, Mengzhou Xia, Alexis Chevalier, Tanya Goyal, Danqi Chen, and Tianyu Gao. 2024. LitSearch: A retrieval benchmark for scientific literature search. In *Proceedings of the 2024 Conference*

on Empirical Methods in Natural Language Processing (EMNLP), pages 15068–15083.

Marwah Alaofi, Paul Thomas, Falk Scholer, and Mark Sanderson. 2024. LLMs can be fooled into labelling a document as relevant: best café near me; this paper is perfectly relevant. In *Proceedings of the 2024 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region (SIGIR-AP)*, pages 32–41.

Negar Arabzadeh and Charles L. A. Clarke. 2025. Benchmarking LLM-based relevance judgment methods. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 3194–3204.

Jaime Arguello, Adam Ferguson, Emery Fine, Bhaskar Mitra, Hamed Zamani, and Fernando Diaz. 2021. Tip of the tongue known-item retrieval: A case study in movie identification. In *Proceedings of the 2021 Conference on Human Information Interaction and Retrieval (CHIIR)*, pages 5–14.

Artificial Analysis. 2026. Artificial analysis search index: Same agent, different search. Launched August 18, 2026; accessed November 3, 2026.

Leif Azzopardi, Maarten de Rijke, and Krisztian Balog. 2007. Building simulated queries for known-item topics: An analysis using six European languages. In *Proceedings of the 30th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 455–462.

Simone Balloccu, Patrícia Schmidtová, Mateusz Lango, and Ondřej Dušek. 2024. Leak, cheat, repeat: Data contamination and evaluation malpractices in closed-source LLMs. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 67–93.

Krisztian Balog, Donald Metzler, and Zhen Qin. 2025. Rankers, judges, and assistants: Towards understanding the interplay of LLMs in information retrieval evaluation. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 3865–3875.

Janek Bevendorff, Matti Wiegmann, Martin Potthast, and Benno Stein. 2024. Is Google getting worse? A longitudinal investigation of SEO spam in search engines. In *Advances in Information Retrieval (ECIR 2024)*, pages 56–71. Springer.

Chris Buckley, Darrin Dimmick, Ian Soboroff, and Ellen M. Voorhees. 2007. Bias and the limits of pooling for large collections. *Information Retrieval*, 10(6):491–508.

Benjamin A. Carterette. 2012. Multiple testing in statistical analysis of systems-based information retrieval experiments. *ACM Transactions on Information Systems*, 30(1):4:1–4:34.

Zijian Chen, Xueguang Ma, Shengyao Zhuang, Ping Nie, Kai Zou, Sahel Sharifymoghaddam, et al. 2026. BrowseComp-Plus: A fair and disentangled evaluation benchmark for deep search agents. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 22349–22370.

Charles L. A. Clarke and Laura Dietz. 2025. LLM-based relevance assessment still can't replace human relevance assessment. In *Proceedings of the International Workshop on Evaluating Information Access (EVIA 2025), a Satellite Workshop of the NTCIR-18 Conference*, pages 1–5.

João Coelho, Jingjie Ning, Jingyuan He, Kangrui Mao, Abhijay Sai Paladugu, Pranav Setlur, et al. 2026. DeepResearchGym: A free, transparent, and reproducible sandbox for deep research. In *Proceedings of the 2026 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR)*, pages 34–43.

Jacob Cohen. 1968. Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit. *Psychological Bulletin*, 70(4):213–220.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186.

Caroline M. Eastman and Bernard J. Jansen. 2003. Coverage, relevance, and ranking: The impact of query operators on web search engine results. *ACM Transactions on Information Systems*, 21(4):383–411.

Bradley Efron and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*, volume 57 of *Monographs on Statistics and Applied Probability*. Chapman & Hall, New York.

Europe PMC Consortium. 2015. Europe PMC: a full-text literature database for the life sciences and platform for innovation. *Nucleic Acids Research*, 43(D1):D1042–D1048.

Guglielmo Faggioli, Laura Dietz, Charles L. A. Clarke, Gianluca Demartini, Matthias Hagen, Claudia Hauff, et al. 2023. Perspectives on large language models for relevance judgment. In *Proceedings of the 2023 ACM SIGIR International Conference on the Theory of Information Retrieval (ICTIR)*, pages 39–50.

Google. 2025. *Search Quality Rater Guidelines: General Guidelines*. Version of September 11, 2025.

Yifan He, To Eun Kim, Fernando Diaz, Jaime Arguello, and Bhaskar Mitra. 2025. Tip of the tongue query elicitation for simulated evaluation. In *Proceedings of the 48th International ACM SIGIR Conference on*

Research and Development in Information Retrieval, pages 3398–3407.

Abe Bohan Hou, Orion Weller, Guanghui Qin, Eugene Yang, Dawn Lawrie, Nils Holzenberger, et al. 2025. CLERC: A dataset for U.S. legal case retrieval and retrieval-augmented analysis generation. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 7913–7928.

Liang Hu, Jianpeng Jiao, Jiashuo Liu, Dongyuan Mutu, Yanle Ren, Zhoufutu Wen, et al. 2026. FinSearchComp: Towards a realistic, expert-level evaluation of financial search and reasoning. In *The Fourteenth International Conference on Learning Representations (ICLR)*.

Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, et al. 2025. LiveCodeBench: Holistic and contamination free evaluation of large language models for code. In *International Conference on Learning Representations (ICLR)*.

Kalervo Järvelin and Jaana Kekäläinen. 2002. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422–446.

Armand Joulin, Edouard Grave, Piotr Bojanowski, Matthijs Douze, Hervé Jégou, and Tomas Mikolov. 2016. FastText.zip: Compressing text classification models. *Preprint*, arXiv:1612.03651.

Armand Joulin, Edouard Grave, Piotr Bojanowski, and Tomas Mikolov. 2017. Bag of tricks for efficient text classification. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pages 427–431.

Evangelos Kanoulas and Javed A. Aslam. 2009. Empirical justification of the gain and discount function for nDCG. In *Proceedings of the 18th ACM Conference on Information and Knowledge Management (CIKM)*, pages 611–620.

Jungo Kasai, Keisuke Sakaguchi, Yoichi Takahashi, Ronan Le Bras, Akari Asai, Xinyan Yu, et al. 2023. RealTime QA: What's the answer right now? In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.

Gabriella Kazai, Bhaskar Mitra, Anlei Dong, Nick Craswell, and Linjun Yang. 2022. Less is less: When are snippets insufficient for human vs machine relevance estimation? In *Advances in Information Retrieval: 44th European Conference on IR Research (ECIR 2022), Part II*, pages 153–162.

M. G. Kendall. 1945. The treatment of ties in ranking problems. *Biometrika*, 33(3):239–251.

Elisabeth Kirsten, Jost Grosse Perdekamp, Qinyuan Wu, Mihir Upadhyay, Krishna P. Gummadi, and Muhammad Bilal Zafar. 2026. Characterizing web search in the age of generative AI. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 10827–10848.

Dirk Lewandowski. 2015. Evaluating the retrieval effectiveness of web search engines using a representative query sample. *Journal of the Association for Information Science and Technology*, 66(9):1763–1775.

Zhuofeng Li, Dongfu Jiang, Xueguang Ma, Haoxiang Zhang, Ping Nie, Yuyu Zhang, Kai Zou, Jianwen Xie, Yu Zhang, and Wenhu Chen. 2026. OpenResearcher: A fully open pipeline for long-horizon deep research trajectory synthesis. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, pages 908–937.

Jimmy Lin. 2009. Is searching full text more effective than searching abstracts? *BMC Bioinformatics*, 10:46.

Grégoire Mialon, Clémentine Fourrier, Craig Swift, Thomas Wolf, Yann LeCun, and Thomas Scialom. 2024. GAIA: a benchmark for general AI assistants. In *International Conference on Learning Representations (ICLR)*.

Jingjie Ning, João Coelho, Yibo Kong, Yunfan Long, Bruno Martins, João Magalhães, Jamie Callan, and Chenyan Xiong. 2026. Agentic search in the wild: Intents and trajectory dynamics from 14m+ real search requests. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1404–1415.

Openbenchmarks. 2026. Web search benchmark. Accessed November 3, 2026.

Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, pages 68772–68802.

Francesca Pezzuti, Ophir Frieder, Fabrizio Silvestri, Sean MacAvaney, and Nicola Tonellotto. 2026. How AI agents search. *PLOS ONE*, 21(10):e0360157.

Oscar Sainz, Jon Ander Campos, Iker García-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. 2023. NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 10776–10787.

Sriram Selvam and Anneswa Ghosh. 2026. Similar accuracy, unequal evidence: Search APIs as decision surfaces for tool-using agents. In *Proceedings of the 1st Workshop on Grounding Language Models: Learning Faithfully and Efficiently (GroundLM 2026)*, pages 606–622.

Ian Soboroff. 2025. Don't use LLMs to make relevance judgments. *Information Retrieval Research*, 1(1):29–46.

Karen Spärck Jones and C. J. van Rijsbergen. 1975. Report on the need for and provision of an 'ideal' information retrieval test collection. British Library Research and Development Report 5266, Computer Laboratory, University of Cambridge.

Nandan Thakur, Jimmy Lin, Sam Havens, Michael Carbin, Omar Khattab, and Andrew Drozdov. 2025. FreshStack: Building realistic benchmarks for evaluating retrieval on technical documents. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.

Paul Thomas, Seth Spielman, Nick Craswell, and Bhaskar Mitra. 2024. Large language models can accurately predict searcher preferences. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1930–1940.

Andrew Turpin, Falk Scholer, Kalervo Järvelin, Mingfang Wu, and J. Shane Culpepper. 2009. Including summaries in system evaluation. In *Proceedings of the 32nd Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 508–515.

Shivani Upadhyay, Ronak Pradeep, Nandan Thakur, Daniel Campos, Nick Craswell, Ian Soboroff, and Jimmy Lin. 2025. A large-scale study of relevance assessments with large language models using UMBRELA. In *Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR)*, pages 358–368.

Denny Vrandečić and Markus Krötzsch. 2014. Wikidata: A free collaborative knowledgebase. *Communications of the ACM*, 57(10):78–85.

Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, et al. 2024. FreshLLMs: Refreshing large language models with search engine augmentation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 13697–13720.

Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, et al. 2025. BrowseComp: A simple yet challenging benchmark for browsing agents. *Preprint*, arXiv:2504.12516.

Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Benjamin Feuer, Siddhartha Jain, et al. 2025. LiveBench: A challenging, contamination-limited LLM benchmark. In *International Conference on Learning Representations (ICLR)*.

Edwin B. Wilson. 1927. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212.

Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V. Le, Mohammad Norouzi, Wolfgang Macherey, et al. 2016. Google's neural machine translation system: Bridging the gap between human and machine translation. *Preprint*, arXiv:1609.08144.

Ori Yoran, Samuel Joseph Amouyal, Chaitanya Malaviya, Ben Bogin, Ofir Press, and Jonathan Berant. 2024. AssistantBench: Can web agents solve realistic and time-consuming tasks? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8938–8968.

Yunfan Zhang, Kathleen McKeown, and Smaranda Muresan. 2026. LiveNewsBench: Evaluating LLM web search capabilities with freshly curated news. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*.

Heng Zhou, Ao Yu, Yuchen Fan, Li Kang, Jianing Shi, Yongting Zhang, et al. 2026a. An automated pipeline for provably retrieval-dependent benchmark construction over dynamic knowledge. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, pages 40633–40648.

Yuqi Zhou, Sunhao Dai, Changle Qu, Liang Pang, Jun Xu, and Ji-Rong Wen. 2026b. Learning to retrieve from agent trajectories. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2544–2555.

Justin Zobel. 1998. How reliable are the results of large-scale information retrieval experiments? In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 307–314.

A Run Protocol and Reproduction

Figure 2 shows the workflow for all five benches.

A.1 Requests and configurations

All engines receive the same query and requested k (Table 2). Each engine processes one request at a time; engines run concurrently. Requests have a 30-second timeout and up to three attempts, with exponential back-off from 1 to 30 seconds on HTTP 429, 5xx and timeouts. We record the error type and message after the final failure. Short result lists score as returned. Snippets have a 2,000-character cap before matching or judging.

The generators and judge use GPT-6 Luna (`openai/gpt-6-luna`) through OpenRouter, temperature 0 and high reasoning effort. Table 3 fixes the search clients and settings. A day costs about 30,700 search requests across 26 configurations and up to 75,000 result judgments.

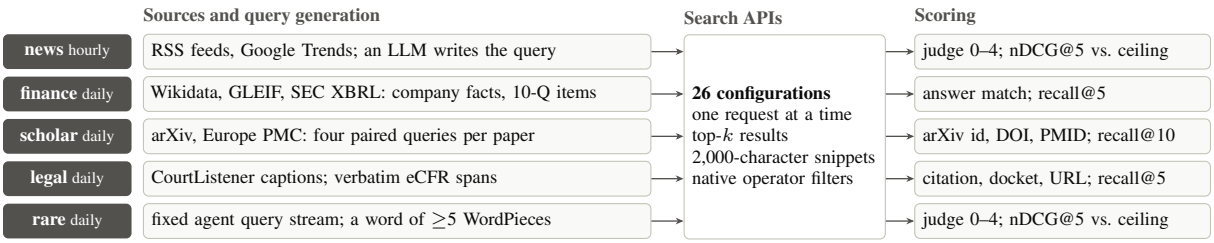


Figure 2: Each query goes to every configuration without its reference answer. The scorer checks the returned title, URL and snippet against that answer, or the judge rates relevance.

Bench	Cadence	Queries/run	k	Metric
news	hourly	20	5	nDCG@5
finance	daily	120	5	recall@5
scholar	daily	420	10	recall@10
legal	daily	40 + 0–20	5	recall@5
rare	daily	100	5	nDCG@5

Table 2: Scholar samples seven papers per domain–age cell, four queries each. Legal adds one caption query per opinion published by the 14 courts.

A.2 Score sources and failure rules

Tables 1, 6 and 8, Figure 1 and the primary paired tests use published `data/history.jsonl` from the `gh-pages` branch. Per-query checks read archived results and import the repository's scoring code. `paper/analysis/manifest.json` lists each result's data, runs, engines and rules.

The code classifies each failed request with `account_side()`. It excludes HTTP 401, 402, 403, 429, 432 and 433, plus HTTP 400 or API errors naming credit, usage limits, quota, billing or an API key. Other failures, including timeouts and server errors, score zero.

The published news and rare scores in the main window count all failures as zero; per-query checks exclude account-side failures. This raises the per-query overall base by up to 6.3 points (Claude search). Eleven of the 12 published scholar runs use a different DOI matcher. Finance, scholar and legal per-query checks nevertheless reproduce every main-window dashboard row. The later-day news and rare scores exclude every failure. These differences matter when comparing the checks with the main table; they do not change the stored table values.

A.3 Rebuilding the results

From `paper/`, run the following on dashboard history retaining rows through 11:53 UTC on 29 October 2026:

```
uv run python \
  analysis/table_rows.py \
  history.jsonl table_results.tex \
  analysis/audit/matched_pairs.json
uv run python \
  analysis/adjacent_pairs.py \
  history.jsonl \
  analysis/audit/adjacent_pairs.json
```

The first command rebuilds the main table and intervals; the second rebuilds the paired tests. Audit scripts, samples and labels are under `paper/analysis/`. Repository and dataset revision hashes are withheld for review. The archive stores ratings but not API publication dates or per-run code revisions, so re-judging is approximate.

A.4 Data and release terms

The rare stream draws from DeepResearchGym (Coelho et al., 2026) search logs (59%, CC BY 4.0; Ning et al., 2026), OpenResearcher trajectories (39%, MIT; Li et al., 2026) and LRAT-Train (1%, no stated license; Zhou et al., 2026b). Source terms govern query texts. Our code is MIT; our CC BY 4.0 covers our results, judgments and metadata, not source text. The release plan removes LRAT queries without permission. They account for 113 of 6,901 filtered queries and 17 of the main window's 1,200 rare query-runs; removing them changes no mean rare nDCG@5 by more than .003 (`rare_drop_lrat.py`).

Filters reject links, common host names and hash, wallet or vehicle ids, and cap near-duplicates. A scan of the 1,100 released rare queries finds no email or HTTP/WWW link, 66 host-like names (61 in `site:`) and four long mixed-case alphanumeric tokens. Queries can name people. A human reads each new query artifact before release; removal requests go to the repository issue tracker. We publish API titles, URLs and snippets as evaluation records under our reading of vendor terms and withdraw an engine's archived outputs on request.

Engine	Endpoint	Variant or non-default parameters
Keenable	api.keenable.ai	default; realtime variant
Exa	api.exa.ai	`type=auto`; instant and lite variants
Google	google.serper.dev (Serper)	default
Bing	searchapi.io (SearchAPI)	engine `bing`
Brave	api.search.brave.com	web search; LLM-context endpoint
Parallel	api.parallel.ai	default processor; turbo variant
Perplexity	api.perplexity.ai	search API; fast variant
Tavily	api.tavily.com	default
Linkup	api.linkup.so	default
TinyFish	api.search.tinyfish.ai	default
Octen	api.octen.ai	default
Ceramic	api.ceramic.ai	default
You.com	ydc-index.io	default
Firecrawl	api.firecrawl.dev	search endpoint, default
Kagi	kagi.com/api/v1	default
Jina	s.jina.ai	default
Context	api.context.dev/v1	default
DeepSeek	api.deepseek.com	`/v1/tools/search`, default
Claude search	api.anthropic.com	`claude-sonnet-5`, `web_search`
ChatGPT search	api.openai.com	`gpt-5.5`, `web_search`, `search_context_size=low`

Table 3: API configurations. Clients live in `src/needle/shared/search/`; Table 11 gives operator mapping.

B Query Filters and Scoring Details

B.1 Query filters

Scholar balances five domains (computer, physical, life, social and health sciences) and three age bands (7 days, 30 days, 1 year). Each paper must pass all four query filters:

- **Title:** at least four words, plus a digit, acronym, hyphenated term or five content tokens.
- **Body keywords:** at least three words and two content tokens absent from title and abstract.
- **Body question:** at least eight words and the same novelty test as body keywords.
- **Tip-of-tongue:** at least twelve words; no quotes, decimal, five-digit number or coined title/abstract term; at most two content tokens shared with the title.

An instrumented run accepts 323 of 572 candidates (56%); domain rates range from .41 to .73. Each domain–age cell takes its first seven accepted papers. The effect of rejecting other papers on engine order is untested (`scholar_rejects.py`).

A CFR query needs at least three words and exactly one verbatim span of two to eight words. It must not contain a citation form (CFR, U.S.C., §, "section n", "title n") or the target part or section number. Only the quoted variant retains quotation marks. News prompts reject evergreen items and request a two-to-six-word query about the event. Table 4 shows sample queries.

B.2 Judge profile and checks

A profile call sees the query and run date. It returns the query type (broad, specific or list), domain, operators, one to three main aspects, auxiliary aspects and a content type (ephemeral, persistent or evergreen). Aspects describe required content, not the answer. Every result uses the same stored profile.

The result call sees that profile, the run date, title, URL, API publication date if available, and snippet. It emits eight checks and a 0–4 rating: integrity, relevance, main-aspect coverage, list completeness, event freshness, staleness, a domain cap and operator compliance. Failed relevance or operator compliance forces 0; failed coverage or completeness caps the rating at 2. Staleness lowers ratings by one or two levels, or to 0 for a wrong year or entity status. A dated-event result about the entity but not the event is capped at 1. Quotes must appear verbatim in the title or snippet; `site:` must match the URL host. The judge must use the result's text and authority, not its own answer knowledge.

Ratings 0–1 mean Fails to Meet, 2 Slightly Meets, 3 Highly Meets and 4 Fully Meets. The scale has no Moderately Meets level, and rating 4 does not need the single-answer intent that the guidelines require for Fully Meets (Google, 2025). Prompts live in `src/needle/shared/prompts`.

Bench	Query	Reference answer or profile
news	Paramount Warner Bros takeover closes 2026	closing date; ephemeral event
finance, fact	nvidia number of employees	42,000, ±15%
finance, money	"servicenow" q3 fiscal 2024 operating income	$418M, ±0.1%
scholar, title	optimally selling heterogeneous objects by quantity	arXiv 2610.36244
scholar, body	gap rent regularity	same paper
legal, CFR	"modify or waive any" county committees authority	7 CFR 1417.2
rare	"Krasnogvardeysky District" Russia	judged; rare word has 7 pieces

Table 4: Sample generated queries. News and rare have no reference answer; the judge profile specifies what a result must contain.

B.3 Ceiling and worked example

For ratings r_i and gains $g(4) = 1$, $g(3) = .667$, $g(2) = .117$, $g(1) = g(0) = 0$,

$$\mathrm{DCG}_e(q) = \sum_{i \le 5} \frac{g(r_i)}{\log_2(i+1)},$$

$$\mathrm{nDCG}_e(q) = \frac{\mathrm{DCG}_e(q)}{\mathrm{IDCG}(q)}.$$

IDCG is the DCG of the top five pooled ratings, keeping the best rating per normalized URL. A bench metric is the mean over scored query-runs; the judged ceiling has mean 1. A shorter returned list gets no padding.

For example, A returns ratings 4 and 2; B returns 3 and 4, with the 4-rated URL shared by both. The pooled ratings are 4, 3, 2, so $\mathrm{IDCG} = 1 + .667/1.585 + .117/2 = 1.479$. A scores $(1 + .117/1.585)/1.479 = .73$; B scores $(.667 + 1/1.585)/1.479 = .88$.

URL normalization removes tracking parameters. Later copies of a URL score zero except on `site:` queries. That exception affects 3 of 1,748 rare `site:` engine–query pairs in the main window and can produce nDCG@5 above 1 because the ceiling still keeps each URL once.

B.4 Finance matching rules

Money must fall within 0.1% of the reference value; headcounts within 15%. A scale word fixes a money figure's scale. A bare figure, including a parenthesized loss, can mean thousands or millions only if the preceding 200 characters contain $, USD, dollar, millions, thousands, revenue, sales, income, loss, earnings, profit or EPS. Years, percentages and multiples do not qualify. Diluted EPS also needs EPS, "per share" or "earnings per" in the result. No money result gets an LLM backstop. Other fields can receive an upgrade, never a revoked hit. A known-item hit counts on any host, even for `site:` queries; this differs from the judged benches' literal operator rule.

The rules live in `src/needle/finance/canon.py` and `src/needle/shared/search/errors.py`. Appendix E checks their error rates.

C Robustness of Engine Comparisons

This appendix tests sampling, missing requests, normalization and time. It uses all active configurations where stated, including variants absent from Table 1. Supporting scripts and outputs are under `paper/analysis/`; `manifest.json` gives their inputs.

C.1 Paired tests and repeated queries

The main window contains 12 scholar, 232 news, 10 finance, 12 rare and 10 legal runs (276 total, 12,353 query-runs). We resample runs 2,000 times, recompute every overall share and take each pair's difference. A 95% interval excludes zero for 188 of 210 pairs and 9 of 20 adjacent pairs. Bonferroni adjustment uses the bootstrap standard deviation and a normal approximation, $z = 3.68$ (`adjacent_pairs.py`).

Table 5 compares sampling units. Of 178 primary adjusted separations, 164 also hold with day-level resampling and matched query support. Marginal interval overlap is not a pairwise test; unresolved pairs do not form groups of equal engines. Full pair lists appear in `audit/adjacent_pairs.json` and `audit/pairs_survive.json`.

Exact query text recurs across runs for 33% of finance, 16% of rare, 11% of news, 5% of scholar and 2% of legal query-runs, counting first occurrences. Day resampling does not cover cross-day repeats; clustering by exact query text (paper on scholar) checks that dependence. Its intervals are narrower than run-level intervals

Check	95%	Adjusted
Run resampling (primary)	188	178
Day resampling	185	174
Query-cluster resampling	188	180
Matched query support	189	171

Table 5: Separated engine pairs out of 210. Adjusted columns use Bonferroni correction. Query clusters use exact text, or the paper for scholar.

for 145 of 210 pairs (`repeat_queries.py`, `cluster_pairs.py`).

C.2 Account-side failures and matched support

Account-side failures cluster by engine and date. Claude search loses 35.9% of requests to exhausted credit, TinyFish 29.1% to rate limits, ChatGPT search 11.0% to credit-related HTTP 429, and Kagi 7.6% to HTTP 401. Jina, Parallel and Exa lose 2.5%, 0.8% and 0.3% to payment errors. The main table excludes these only on finance, scholar and legal, leaving TinyFish without 27.9% of its query-runs and Claude search without 16.9%. Other failures affect 2.4% of Jina requests, 0.4% of Bing requests and at most 0.1% for other engines (`failure_profile.py`).

Scoring only answered requests raises Claude search from 45.1 to 51.4, rank 16 to 13; no other share moves more than 0.9 points. To remove unequal query support, we also drop a query-run for every engine if any engine has an account-side failure. This retains 4,048 of 12,313 query-runs with available artifacts (230 of 232 news runs and every other run). Table 6 reports changes from the per-query base, which already excludes account-side failures on news and rare. The matched overall order has $\tau_b = .89$ against the main table; Keenable and Exa stay first and second, Tavily and Ceramic last (`matched_pairs.py`). We keep the larger, unequal-support estimate as primary and show both.

Exclusion assumes failed requests would score like answered ones. Scoring each excluded request as zero or at the ceiling bounds its effect (Table 7). Ranks 1–10 remain fixed under both extremes. These bounds also include finance queries dropped after a backstop error, not the judged query-runs dropped for all engines (`exclusion_bounds.py`).

C.3 Ceiling, gains and bench weights

Removing the scored engine from the pool can only raise its share, since the ceiling can only fall. Across all five benches and the 21 main engines, the largest overall change is 0.8 points; only Context and DeepSeek swap places (`loo_all.py`).

On 230 news runs, 6 of 4,522 fully judged query-runs have an empty ceiling; on rare, 147 of 1,189 do (12.4%). Failed judgments also drop 59 news and 11 rare query-runs. Counting empty-ceiling queries as zero for both engine and ceiling lowers absolute means but moves no bench share by more than .002 (`ceiling_drop.py`). The main table's nDCG@5 cells are thus conditional means.

Alternative news gains yield Kendall τ_b .95 for linear or exponential gains and .98 for binary relevance at rating ≥ 3; stricter or looser binary cutoffs give .81–.87. Averaging absolute bench metrics instead of ceiling shares gives overall $\tau_b = .99$, swapping Context and DeepSeek (`operators/sensitivity.py`).

Keenable's 4.7-point lead over Exa comprises scholar +3.7, rare +0.6, legal +0.5, news +0.2 and finance -0.4. Dropping any one bench retains the leader but gives leads from 1.2 points (without scholar) to 6.4 (without finance). On scholar, the recall lead is +.38 for 7-day papers, +.10 for 30-day papers and +.03 for one-year papers. It is positive in four domains but $-.10$ in health sciences, and +.01 on titles versus +.17 on body keywords and +.33 on body questions. This is mainly a recent-paper and full-text lead, not a uniform one (`bench_weights.py`, `scholar_strata.py`). Table 8 gives all query types.

C.4 Time, dispatch and later runs

Over eight weeks, adjacent engine orders have τ_b .81–.94. The 15 configurations present in every week give .85–.94; fixing the news pool to those 15 changes no sampled weekly order. Filtering to weeks' runs where Claude search loses at most 10% of requests still lowers its share from 58.3 to 52.2. The rest of its plotted decline reflects our credit lapse (`weekly_tau.py`, `fixed_pool_weekly.py`, `claude_weekly.py`).

In 249 news runs with timestamps, first-to-last dispatch of the same query has median 6.5 minutes. Without Jina, Claude search and ChatGPT search it is 31 seconds. Keeping only pairs dispatched

Engine	Share	95% CI	Δ matched	95% CI	Sch.	News	Fin.	Rare	Legal
Keenable	73.2	72.3–74.1	+1.6	+0.4, +2.7	.811	.609	.909	.508	.821
Exa	68.5	67.4–69.4	−1.1	−2.3, +0.2	.625	.599	.929	.476	.795
Perplexity	66.9	65.3–68.5	−0.7	−2.5, +1.1	.641	.640	.902	.468	.694
Parallel	64.9	63.6–66.1	−2.1	−3.5, −0.7	.651	.535	.914	.387	.757
Brave (LLM ctx)	64.6	63.4–65.6	+2.5	+1.1, +3.9	.591	.642	.865	.454	.678
Linkup	57.6	56.1–58.9	+1.7	−0.1, +3.8	.521	.567	.803	.400	.588
Brave	57.3	56.2–58.4	+1.8	+0.2, +3.2	.541	.557	.775	.354	.640
Context	57.1	55.4–58.6	−0.9	−3.0, +1.1	.664	.469	.788	.339	.594
DeepSeek	56.9	55.8–57.9	+1.1	−0.6, +2.9	.301	.635	.824	.387	.696
You.com	55.8	54.7–56.9	+1.6	+0.3, +2.9	.523	.553	.752	.343	.620
ChatGPT search	51.5	49.4–53.1	−0.5	−2.0, +0.8	.333	.340	.816	.366	.718
Kagi	51.3	49.9–52.6	+0.1	−0.8, +1.2	.621	.301	.734	.275	.635
Firecrawl	50.7	49.2–52.0	−0.5	−1.9, +0.9	.541	.286	.826	.265	.616
Octen	50.2	48.8–51.7	−1.1	−2.7, +0.6	.340	.511	.735	.322	.604
Google (Serper)	47.7	46.8–48.6	−0.3	−1.5, +0.9	.527	.262	.743	.255	.600
Claude search	45.1	42.7–47.6	+1.2	−0.4, +2.9	.553	.276	.690	.195	.541
TinyFish	44.5	43.0–45.9	−0.4	−2.0, +1.2	.476	.256	.736	.220	.537
Bing (SearchAPI)	44.1	41.1–46.8	+4.1	+1.5, +7.5	.461	.288	.680	.173	.604
Jina	43.8	42.4–45.1	−0.5	−2.2, +1.2	.545	.210	.731	.163	.539
Tavily	42.6	41.5–43.7	−0.1	−1.2, +1.3	.336	.303	.834	.284	.372
Ceramic	14.7	13.9–15.7	−0.1	−1.1, +1.2	.002	.040	.444	.109	.141

Table 6: Overall shares in percent; bench shares as fractions. Δ matched is the change in percentage points when all engines use the same 4,048 query-runs, with a run-resampled 95% interval. Both terms of that change use per-query scoring; its base exceeds the Share column by up to 6.3 points (Claude search) because of failure handling (Appendix A).

Engine	Overall share bounds, %	Rank range
Claude search	34.7–52.0	11–20
TinyFish	37.6–52.0	11–20
Kagi	47.5–53.2	11–15
ChatGPT search	48.2–54.5	11–14
Jina	42.9–44.4	18–19

Table 7: Bounds from scoring excluded requests as zero or at the ceiling. No other engine has such exclusions in the main table.

within five minutes of the first engine retains 96% of scored pairs, leaves the engine order unchanged and moves no mean nDCG@5 by more than .010 (`dispatch_gap.py`, `lag_subset.py`).

The later-day check covers 28 news runs and one run of each other bench, from 30 October to early 31 October 2026: 1,224 query-runs per engine, 19 engines with all five benches. Claude search lacks finance after a credit lapse; all Kagi requests fail. The overall order has $\tau_b = .86$ against the main table, with the same five leaders and Ceramic last, but Perplexity leads Keenable 72.4 to 71.3. Their single scholar and legal runs explain the change and stay within the main window's run-to-run ranges. Google and Jina carry redirect URLs on about half of results, which hides identifiers from URL matchers. One day checks variation; it does not establish a new ranking (`later_day.py`).

D Judge and Identifier Checks

D.1 Blinded raters

We sample 100 news and 100 rare queries from 17–29 October runs with at least 19 engines, taking six distinct top-five URLs per query (1,200 results). Two Claude Opus 5.5 raters see the run date, query, URL, title, snippet and first 3,000 page characters, with engine names hidden. Each uses its own random order in 24 sessions of 50 items. Gemini 4 Argon and Grok 4.7 grade the same items, one item per call at temperature 0, under the same written guideline (`audit/raters/`). No human checks the ratings.

Raters agree with one another more than with the production judge (Table 9) and assign higher mean ratings: 1.87–1.97 versus 1.60. Pages are missing or broken for 128 items. Because the Claude raters see the page while recording both grades, a separate staged pass grades 400 items from snippets alone. Its grades differ from page grades on 25% of items (95% interval 21–29%), with a mean page-minus-snippet change of +.05 (−.01 to +.11). This pooled check does not rule out engine-specific page effects (`staged_page.py`).

D.2 Complete lists and a second judge

Six sampled results per query cannot yield a list metric. Gemini and Grok therefore also grade

Engine	title	body keyword	body question	tip of tongue	query-runs
ceiling (pooled)	*1*	*.888*	*.938*	*.767*	*12,020*
Parallel	**.981**	.453	.642	.278	12,020
Kagi	.962	.674	.640	.091	11,599
Bing (SearchAPI)	.903	.402	.411	.060	12,020
Parallel Turbo	.894	.208	.294	.110	12,020
Brave (LLM ctx)	.893	.440	.620	.194	12,020
ChatGPT search	.886	.083	.192	.118	11,016
You.com	.883	.370	.593	.129	12,020
Brave	.881	.383	.613	.167	12,020
Keenable	.879	**.721**	**.800**	**.510**	12,020
Linkup	.878	.357	.575	.130	7,898
Keenable realtime	.877	.709	.796	.509	12,020
Perplexity	.865	.585	.668	.309	11,636
Jina	.862	.629	.470	.030	11,716
Perplexity Fast	.860	.574	.666	.305	11,636
Claude search	.860	.427	.620	.182	10,400
Context	.856	.628	.516	.111	12,020
Exa Lite	.848	.371	.305	.219	420
Firecrawl	.836	.636	.472	.028	12,020
Google (Serper)	.834	.601	.457	.030	12,020
Exa	.805	.531	.462	.345	12,020
Tavily	.795	.279	.144	.033	12,020
TinyFish	.769	.563	.440	.018	3,472
Exa Instant	.713	.409	.374	.273	12,020
Octen	.660	.063	.147	.157	12,020
DeepSeek	.608	.183	.184	.109	4,820
Ceramic	.011	.000	.000	.000	12,007

Table 8: Scholar recall@10 by query type, 31 runs, 29 September–28 October 2026. The ceiling has 3,005 query-runs per type; the last column sums all four types. Counts fall where an engine misses runs or loses queries. Bold marks the best configuration per type (`bucket_table.py`).

100 complete pools (50 news, 50 rare) from the main window: all 21 engines' top-five lists, 5,698 unique results, snippets only. Recomputed pooled nDCG@5 ranks engines at Spearman .94 against the judge for each rater (95% intervals .89–.97 and .87–.96). Mean absolute share changes are .07 and .06. All three retain the same top five, in different orders. Keenable–Exa is unresolved under each: +.035 ($-.024$ to +.091), +.037 ($-.016$ to +.089) and .000 ($-.053$ to +.053) for judge, Gemini and Grok (`list_analysis.py`, `list_pair_diffs.py`).

A second judge, Claude Opus 5.5, uses the production prompts on eight queries per bench (803 news and 566 rare results). Quadratic κ with the production judge is .79 on news and .90 on rare; engine-share Spearman correlations are .90 and .87. API publication dates are not archived, so this judge lacks that input. Re-running the production judge without dates gives $\kappa = .80$ against its original news grades and .97 on rare: the date effect is as large as the news model effect. Both

Quadratic-weighted κ	News	Rare	All
Claude A vs. B	.94	.94	.94
Judge vs. Claude consensus	.72	.80	.77
Gemini vs. consensus	.91	.92	.92
Grok vs. consensus	.90	.88	.90
Gemini vs. judge	.74	.81	.78
Grok vs. judge	.75	.81	.79

Table 9: Agreement on 1,200 blinded results, 600 per bench. Consensus is the rounded mean of the two Claude page grades. The judge–consensus 95% query-bootstrap interval is .74–.81.

judges also share the stored profile. A separate LLM profile check finds 13 of 16 aligned in intent, two narrower and one broader, with no wrong intent or answer leak (`rejudge_analysis.py`, `audit/profiles/`). Eight queries per bench give only a small-sample check.

D.3 Snippet length and literal quotes

Median news snippets range from 117 characters (Bing) to the 2,000-character cap (Jina), with rank correlation .34 against news share. Rejudging the 803 news results at a common 200-character cap gives engine-order $\tau_b = .73$ (95% interval .40–.84) against the second judge's full-snippet order. Mean absolute share change is .084; Exa and Keenable lose, while Bing and Google gain. The leader and bottom two remain, but the sample does not separate the top two at either length (`snippet_length.py`, `trunc_ci.py`). Snippet length is part of the API response and can materially affect the score.

Of 1,200 rare query-runs, 504 contain quotes; 30 wrap the whole text as `query: "..."`. Dropping wrappers moves no rare share more than .002. On unquoted queries, Exa, Perplexity and Brave (LLM ctx) lead; Keenable is fourth. Using only unquoted rare scores leaves overall $\tau_b = .96$ and a 3.3-point Keenable lead over Exa. Literal quote compliance therefore affects the rare ranking. Among 212 quoted results with page text, 74 contain the phrase in the title or snippet and only eight more on the page alone (`rare_quoted.py`, `unquoted_overall.py`, `quoted_on_page.py`).

D.4 Identifier hits and misses

In three scholar runs, 95.1% of 13,311 hits carry the id in their own URL. Across ten legal runs, 94.0% of CFR hits carry the section in the URL; 95.4% sit on primary legal hosts. Caselaw is less direct:

a blinded LLM audit of 100 hits finds 80 opinion or docket pages, 14 other pages about the same case, five unclear and one wrong case. Multiplying each engine's caselaw hits by the pooled target-page fraction .80 lowers legal recall .01–.04 and gives $\tau_b = .96$. Engine-specific corrections give .78, but rest on only four sampled hits per configuration. This audit cannot rule out engine-specific error (`legal_target_only.py`).

In the other direction, an LLM finds the target among the top ten in 11 of 100 sampled scholar body-query misses (95% interval .06–.19), under URLs or snippets without the id. Known-item recall can thus miss returned targets as well as credit related pages.

E Finance Matcher Checks

Finance checks use 24 daily runs from 29 September–28 October 2026, 21 engines and 56,760 engine–query pairs without account-side failures. Of these, 13,149 ask for money and 4,393 for headcounts. All judgments below are by LLMs, not humans.

E.1 Bands and precision

Table 10 varies the match bands without table reading or an LLM backstop. A 2% money band adds nine recall points over 0.1%, but admits more wrong periods and related line items.

Money band	0	**.1%**	.5%	1%	2%	5%	10%
recall@5	.363	**.447**	.498	.517	.541	.583	.628
τ_b vs. .1%	.97	1	.95	.94	.98	.91	.88
Headcount band	0	1%	5%	**15%**	25%	50%	
recall@5	.794	.811	.863	**.915**	.919	.935	
τ_b vs. 15%	.73	.77	.81	1	.93	.79	

Table 10: Deterministic recall and engine-order agreement by tolerance over 24 finance runs. Bold marks the production bands (`band_sweep.py`).

A stratified audit of 399 results includes 140 money hits within 0.1%, 80 results 0.1–2% from the answer, 50 headcount hits, 59 other-field hits and 70 backstop upgrades. Claude Opus 5.5 checks the company, period and line item, allowing reporting rounding. Rejected deterministic hits receive a second pass with the full 2,000-character input; rejected backstop upgrades do not. Weighted by hit frequency, precision without table reading is .98 overall and .93 on money. Money precision falls to .88 under a 2% band. Backstop upgrades have precision .36 on money versus .95 on 20 non-money results; production therefore uses no money backstop (`aggregate_audit.py`).

E.2 Table figures and missed answers

Of 80 sampled bare table figures or parenthesized losses equal to the answer at a thousand/million scale, 73 state the right answer. The production money-cue rule accepts 73 and confirms 71. In 100 sampled money query-runs missed even at a 2% band without table reading, an LLM finds the answer in 29. Table reading recovers 19 of those and falsely accepts one other query; 10 of the remaining 80 still contain the answer.

Applying the pooled false-hit and missed-answer rates by field to the main window changes finance recall from $-.025$ (Keenable) to +.021 (Ceramic). It compresses the finance spread from .481 to .443; Perplexity narrowly passes Keenable on finance, and ChatGPT search passes DeepSeek. Overall $\tau_b = .99$, with Kagi passing ChatGPT search and Keenable's lead over Exa at 4.5 points. These are error estimates, not new gold labels. The miss sample cannot estimate each engine's error rate (`finance_bias_table1.py`).

E.3 Fresh-data check

We generate 307 unseen queries (146 money, 161 other) and run them through 17 configurations on 31 October. Claude search lacks credit; Google and Jina return redirect URLs. An LLM grades 335 table-reading hits, 100 exact-band money hits and 50 other-field hits, using at least the full 2,000-character input. Each item is the first matching result in its top five, so the audit checks query-level recall decisions (`fresh_finance.py`, `audit/fresh/`).

Exact-band precision is .93. Of 335 table-reading hits, 235 confirm the right figure and period (.70, Wilson interval Wilson, 1927, .65–.75), 95 omit period labels and five contradict the answer. Precision is .70 if unresolved hits count as wrong, .99 if only contradictions do. Confirmation rates vary: Exa 42/44 and Perplexity 27/31, versus Keenable 14/20, Parallel 36/48, Tavily 12/24 and Firecrawl 13/43. The rule-consistent precision of other-field hits is .90 (`fresh_finance_labels.py`).

Using each engine's fresh unconfirmed fraction to discount its main-window table-reading hits lowers finance recall by up to 6.0 points (Firecrawl); ranks 1–10 overall stay fixed, with $\tau_b = .98$. Drop-

ping table reading entirely lowers finance recall by 0.1–7.5 points and the ceiling from .993 to .988, with finance $\tau_b = .84$ and overall .97; overall ranks 1–3 stay fixed (`reading_correction.py`, `no_table_reading.py`).

Fresh-run finance shares agree with the main column at $\tau_b = .80$, with a mean absolute difference .05 and the same five leading engines. Money queries make up half this sample versus 23% in production, and it lacks a non-money backstop. This checks direction, not the table's exact values.

F Query Operators

F.1 Mapping and compliance

Clients parse `site:`, `after:` and `before:` and use native filters where available (Table 11). Quoted phrases stay in the query text. Google, Bing and Jina receive the raw query.

On finance and legal `site:` queries, 16 of 21 engines return only the requested host. Legal compliance falls to .98 for Claude, .99 for Google, .86 for Bing, .09 for Kagi and .03 for Ceramic. Kagi's lens parameter does not enforce a restriction; Ceramic takes only bare query text. Counting only matching-host hits lowers pooled caption recall from .578 to .527, CFR from .660 to .642 and filing from .654 to .652. Only Kagi and Ceramic move by more than .01. The archive lacks publication dates, so date compliance cannot be checked (`site_compliance.py`, `compliant_hits.py`).

F.2 Paired operator test

Production assigns variants to different targets. To isolate operator effects, we take 80 filing, 80 caselaw and 80 CFR targets, remove operators and quotes, and create quoted, site-restricted and date-restricted variants of each. Dates bracket filing or decision by 45 days. Variants run next to one another through 18 configurations on 30 October, with no Kagi, You.com or DeepSeek (`paired_build.py`, `paired_analysis.py`). A reference-id hit counts on any host, so site losses measure recall, not compliance.

On filings, quotes add .10 mean recall and `site:sec.gov` adds .35; a date window changes it by $-.02$. On captions, `site:courtlistener.com` lowers recall by .36, with intervals below zero for 15 of 18 engines; quotes cost .05 and dates .11. On CFR, quotes add .18 and `site:law.cornell.edu` adds .25. This test does not separate coverage, ranking and filter implementation.

Google and Jina return redirect URLs on 34–43% of results; Context does so on 15% of finance results. Account-side failures remove the target from an engine's difference; other failures score zero. Keenable answers 229/320 finance requests after a missing key triggers public rate limits. Claude has only 29–30 caption targets and no CFR cell; rate limits also reduce Tavily and TinyFish support. These missing targets can bias the estimated operator effect beyond the intervals.

Removing engines with redirects on more than 1% of results leaves the caption site loss at .35, filing site gain at .32, and CFR gains at .14 (quotes) and .20 (site). Among the 14 engines with no failed caption request, the caption site loss remains .32.

Engine	`site:`	`after:`/`before:`
Brave, Brave (LLM ctx)	in query text	freshness range
Keenable	in query text	published-date tokens
Exa	includeDomains	start/endPublishedDate
Tavily, Octen, TinyFish	include_domains	start/end dates
Linkup	includeDomains	from/toDate
Context	includeDomains	freshness (after only)
Perplexity	domain filter	date filters
Parallel	include_domains	after_date only
Kagi	lens sites	after/before filters
You.com	include_domains	freshness range
Firecrawl	in query text	cd_min/cd_max
DeepSeek	site: OR site:	appended to text
ChatGPT, Claude	allowed_domains	in prompt text
Ceramic	dropped	dropped
Google, Bing, Jina	raw string	raw string

Table 11: Operator mapping by API. Ceramic strips operators; its raw query interface returns no results on all 20 operator test queries.

	SEC filings (80 targets)				Caselaw captions (80 targets)				CFR sections (80 targets)		
Engine	plain	quoted	`site:`	dates	plain	quoted	`site:`	dates	plain	quoted	`site:`
Perplexity	.66	$+.06^{\dagger}$	$+.07^{\dagger}$	$+.03^{\dagger}$	.97	$-.03^{\dagger}$	$-.07$	$0^{\dagger}$	.64	$+.14$	$+.11^{\dagger}$
Claude search	.54	$+.01^{\dagger}$	$+.20$	$+.09^{\dagger}$	.97	$-.10^{\dagger}$	$-.30$	$0^{\dagger}$	–	–	–
Parallel	.53	$+.01^{\dagger}$	$+.29$	$+.06^{\dagger}$	.96	$0^{\dagger}$	$-.14$	$+.01^{\dagger}$	.70	$+.09$	$-.01^{\dagger}$
Keenable	.52	$+.12$	$+.28$	$-.15$	.99	$-.09$	$-.03^{\dagger}$	$-.03^{\dagger}$	.64	$+.09$	$+.25$
Exa	.51	$+.16$	$+.28$	$+.14$	.96	$-.03^{\dagger}$	$-.55$	$0^{\dagger}$	.60	$+.14$	$+.21$
ChatGPT search	.45	$+.05^{\dagger}$	$+.25$	$+.05^{\dagger}$	.95	$-.25$	$-.88$	$-.03^{\dagger}$	.56	$+.15$	$+.16$
Brave (LLM ctx)	.29	$+.11$	$+.38$	$-.15$	.96	$0^{\dagger}$	$-.03^{\dagger}$	$-.23$	.53	$+.05$	$+.09^{\dagger}$
Bing (SearchAPI)	.29	$+.01^{\dagger}$	$+.39$	$+.01^{\dagger}$	.91	$-.06^{\dagger}$	$-.62$	$-.06^{\dagger}$	.51	$+.11$	$+.16$
Linkup	.26	$-.01^{\dagger}$	$+.54$	$-.16$	.93	$-.01^{\dagger}$	$-.25$	$-.33$	.45	$0^{\dagger}$	$+.14$
Brave	.23	$+.10$	$+.42$	$-.06^{\dagger}$	.97	$-.16$	$-.06$	$-.33$	.55	$0^{\dagger}$	$+.06^{\dagger}$
Firecrawl	.15	$+.30$	$+.62$	$+.04^{\dagger}$	.89	$-.03^{\dagger}$	$-.65$	$-.01^{\dagger}$	.31	$+.39$	$+.56$
Jina	.15	$+.09^{\dagger}$	$+.49$	$-.07^{\dagger}$	.61	$+.01^{\dagger}$	$-.28$	$-.16$	.25	$+.44$	$+.60$
TinyFish	.12	$+.19$	$+.47$	$-.04^{\dagger}$	.88	$-.14^{\dagger}$	$-.68$	$-.18^{\dagger}$	.21	$+.47$	$+.61$
Tavily	.12	$+.04^{\dagger}$	$+.39$	$-.07^{\dagger}$	.84	$+.04^{\dagger}$	$-.35$	$-.25$	.55	$0^{\dagger}$	$+.07^{\dagger}$
Context	.11	$+.23$	$+.44$	$+.03^{\dagger}$	.82	$+.04^{\dagger}$	$-.61$	$-.03^{\dagger}$	.30	$+.38$	$+.55$
Google (Serper)	.07	$+.24$	$+.56$	$-.03^{\dagger}$	.86	$-.10^{\dagger}$	$-.64$	$-.35$	.26	$+.47$	$+.56$
Octen	.07	$0^{\dagger}$	$+.24$	$0^{\dagger}$	.69	$+.04^{\dagger}$	$-.41$	$-.06^{\dagger}$	.56	$+.07^{\dagger}$	$+.09^{\dagger}$
Ceramic	.00	$0^{\dagger}$	$0^{\dagger}$	$0^{\dagger}$	.19	$-.01^{\dagger}$	$0^{\dagger}$	$0^{\dagger}$	.14	$0^{\dagger}$	$0^{\dagger}$

Table 12: Plain recall@5 and paired changes on the same targets, 30 October 2026. † marks a target-resampled 95% interval containing zero; other changes exclude zero. Redirect URLs affect Google, Jina and Context (finance); failures reduce support for some engines, as noted in the text.